

Trivikram Satharasi

(352) 709 9562 | vikram.satharasi@gmail.com | [Google Scholar](#) | [Github](#)

EDUCATION

Indian Institute of Technology Madras

Bachelor of Technology in Aerospace Engineering

Chennai, India

Nov. 2020 – May 2024

- GPA: 3.93/4.00 (WES Scale) || 9.03/10.00
- Ranked in the top 5 of the graduating class

University of Florida

Master of Science in Aerospace Engineering

Gainesville, FL, USA

Aug. 2024 – May 2026

- GPA: 3.93/4.00
- Advisor: Dr. Rushikesh Kamalapurkar

PUBLICATIONS

1. **T. Satharasi**, T. E. Ogri, M. Qureshi, K. Volle, and R. Kamalapurkar, *Safe Adaptive Feedback Control via Barrier States*, **accepted** for presentation in the *IFAC World Congress 2026* and publication in *IFAC-PapersOnLine*, 2026. arXiv:2601.00476.
2. **T. Satharasi**, T. E. Ogri, M. Qureshi, K. Volle, and R. Kamalapurkar, *Adaptive Control with Sparse Identification of Nonlinear Dynamics*, **accepted** for presentation and publication in the *Proceedings of the IEEE Conference on Decision and Control*, 2026. arXiv:2604.06338.

In Review

1. M. Qureshi, **T. Satharasi**, T. E. Ogri, K. Volle, and R. Kamalapurkar, *A Hough Transform Approach to Safety-Aware Scalar Field Mapping Using Gaussian Processes*, **under review** in *IEEE Trans. Control Sys. Tech.*, 2026. arXiv:2604.20799. (Submitted: March 2026)
2. M. Qureshi, T. E. Ogri, **T. Satharasi**, K. Volle, and R. Kamalapurkar, *Online Input Bias Correction in Gaussian Process Regression*, under review in *IEEE Robotics and Automation Letters*, 2026. (Submitted: May 2026, Revised: August 2026)
3. T. E. Ogri, **T. Satharasi**, M. Qureshi, K. Volle, and R. Kamalapurkar, *On sparsity and directional forgetting in adaptive control*, manuscript in review for potential publication in the *IEEE Transactions on Automatic Control*. (Submitted: August 2026)

In Preparation

1. T. E. Ogri, **T. Satharasi**, M. Qureshi, K. Volle, and R. Kamalapurkar, *Safe Output-Feedback Control via SDREs and Barrier States*, manuscript in preparation, for submission to *IEEE Robotics and Automation Letters*.
2. **T. Satharasi**, T. E. Ogri, M. Qureshi, K. Volle, and R. Kamalapurkar, *Directional Forgetting beyond Constant Subspace Excitation*, **manuscript in preparation** for submission to *IEEE Transactions of Automatic Control*, 2027.

RESEARCH AND TEACHING EXPERIENCE

Graduate Research Assistant

Under Prof. Rushikesh Kamalapurkar at University of Florida

August 2024 – May 2026

Gainesville, FL

- Designed and implemented a reusable ROS 2 navigation framework for nonholonomic ground robots, enabling experimental validation of scalar-field mapping algorithms and supporting 4 journal publications.
- Developed and validated obstacle-avoidance controllers under partial-state feedback through simulation and quadrotor experiments, contributing to a scientific letter manuscript.
- Developed reinforcement learning-based control policies for safe navigation under uncertain dynamics while enforcing obstacle-avoidance constraints.
- Mentored a high school intern in experimental setup using Raspberry Pi, Ubuntu, and ROS 2.

Graduate Teaching Assistant

August 2025 – December 2025

Under Prof. Amor A. Menezes at the University of Florida

Gainesville, FL

- Supported a graduate Control Systems Theory course by grading coursework in Canvas, providing timely assignment feedback, and clarifying course material for students.
- Served as a teaching assistant for an undergraduate Introduction to Aerospace Engineering course, administering quizzes, grading student submissions, providing feedback, and answering questions after class.

Undergraduate Research Assistant

January 2022 – January 2024

National Center for Combustion Research and Development

IIT Madras, Chennai, India

- Created and Fabricated 3D door models in SolidWorks designed for easy opening and closing
- Setup and assisted in Optical Flow Measurements during Wind Tunnel Experiments.
- Designed and implemented a PID controller for controlling the air speed in an accelerating flow based wind tunnel

PROJECTS

Safety - Critical Control in Non Linear Systems

January 2025 – Present

- *Python, Matlab, Gazebo, ROS2, Docker, Hardware, Experiment Setup, Theoretical Control Theory*
- Developed a Safety-Critical Regulation Controller for uncertain and output feedback systems. Furthermore, Lyapunov Theory was used to theoretically guarantee safety and stability.
- Empirically validated the designed controllers by simulating them in MATLAB, Python, and Gazebo across various systems.
- Empirically validated the designed controllers by implementing them on holonomic robots and quadcopters using ROS2.
- Performed Literature Review, implementing foundational work, and performed comparative studies of these controllers with the developed controllers.

Online Adaptive Control with non smooth ℓ_1 regularization

January 2025 – Present

- *Python, Matlab, Non-smooth Lyapunov Theory*
- Developed Novel foundational online adaptive control techniques which use sparsity-inducing ℓ_1 regularization to reduce the number of active basis functions.
- Empirically validated the designed controllers by simulating them in MATLAB and Python, showing great performance increases in a shorter time.
- Performed Literature Review, implementing foundational work, and performed comparative studies of these controllers with the developed controllers in an offline setting.
- Led the preparation of the manuscript for publication and presentation in *IEEE CDC 2026*, and assisted in the preparation of a *IEEE Trans. Automatic Control* manuscript developing novel adaptive control technique using Memory Regressor Extension.

Scalar Field Mapping using Gaussian Process and Hough Transform

January 2025 – May 2026

- *Python, Matlab, ROS2, Gaussian Process, Machine Learning, Raspberry PI, Sensor integration*
- Developed ROS2 workflow for waypoint navigation for non-holonomic ground robots
- Empirically validated the developed method using hardware experiments to map the intensity of light while avoiding the high intensity regions as much as possible
- Developed ROS2 workflows which have become a common starting point in the laboratory for experiments with non-holonomic robots, assisting in about 4 – 5 manuscripts
- Assisted the preparation of the manuscript for publication in *IEEE Trans. Cont. Sys. Tech.*

RESERACH VOLUNTEERING

Reviewer: IEEE Transactions of Automation Science, Springer Nature, IEEE Control Systems Society

AWARDS AND HONORS

1. **64th Foundation Freeship Scholarship**, Indian Institute of Technology Madras, awarding a year's tuition waiver for academic performance.
2. **University of Florida Academic Achievement Award**, University of Florida
3. **IFAC Foundation Travel Grant**, International Federation of Automatic Control, awarding a registration fee waiver to attend the IFAC Conference in Busan.
4. **Member, Phi Kappa Phi Honor Society**, University of Florida, for Interdisciplinary Academic Criterion

EXTRACURRICULARS

1. **Student Mentor**, SAATHI Mentorship Program, IIT Madras (August 2021 - May 2022)
2. **Student Representative**, Aerospace Engineering Department Committee, IIT Madras (August 2021 - May 2024)
3. **Volunteer**, National Service Scheme, IIT Madras (August 2020 - January 2024)
4. **Operations Lead**, Aerospace Engineering Association, IIT Madras (August 2022 - May 2024)